

ASR Shootout - Intern Assessment

Context

You're joining a team that builds voice and telephony infrastructure for a blue-collar hiring platform operating across India. Candidates interact with the platform via phone calls and WhatsApp voice notes - often in Hindi, Hinglish, or regional languages, in noisy environments, over low-bandwidth connections.

Your job is to evaluate the current landscape of Automatic Speech Recognition (ASR) for this stack.

The Task

Benchmark ASR systems for Indian conversational speech and present your findings.

That's it. Everything else - what models you pick, what metrics you measure, how you test, what you conclude - is up to you.

Datasets

Required: Record Your Own

Record **20 audio samples** of Bangalore locality names spoken naturally - the way a candidate on a phone call would say them. Think: "Haan, main Koramangala mein rehta hoon", not a clean studio reading of "Koramangala."

Here are the 20 localities:

Koramangala, Indiranagar, Whitefield, Electronic City, Marathahalli, Jayanagar, Rajajinagar, Hebbal, Yelahanka, Banashankari, HSR Layout, BTM Layout, Majestic, Silk Board, Bellandur, Sarjapur, Bommanahalli, KR Puram, Peenya, Yeshwanthpur, Byatarayanapura, Kadugondanahalli, Hesaraghatta, Chikkabanavara, Rajarajeshwarinagar, Kothanur Dinne, Thanisandra, Doddanekundi, Kengeri Upanagara, and Thalaghattapura.

Recording guidelines:

- Record each locality in a natural conversational sentence (Hindi, Hinglish, or Kannada - your choice)
- Vary the conditions - some from a quiet room, some with street/traffic noise, some over a phone call, some whispered or rushed
- Use your phone mic, not a studio setup - we want real-world audio quality
- Save as individual audio files with clear naming

This is your **primary evaluation dataset**. You are testing whether ASR systems can correctly capture locality names - the kind of entity extraction that matters in a hiring platform where a candidate says where they live.

Optional: Open-Source Datasets

You may additionally use any of these to broaden your evaluation. Not required, but strong submissions will test beyond just their own recordings.

Dataset	What it is	Link
Kathbath	Hindi conversational speech (read + spontaneous), multiple speakers, varied accents	AI4Bharat/Kathbath
Mozilla Common Voice (Hindi)	Crowd-sourced Hindi speech, wide demographic range, variable recording quality	commonvoice.mozilla.org
FLEURS (Hindi)	Multi-speaker Hindi from diverse regions, clean-ish but natural	HuggingFace
IndicVoices	Large-scale Indic speech across multiple languages and domains	AI4Bharat/IndicVoices
MUCS 2021	Hindi-English code-switched speech	Microsoft MUCS

If you find other relevant datasets, feel free to use them. Just tell us why.

Baseline: Deepgram

Deepgram is your baseline. Run all 20 recordings through Deepgram's API and report its performance. Every other model you benchmark should be compared against Deepgram's numbers.

Deepgram offers a free tier that should be sufficient for this task. Sign up at deepgram.com.

Beyond Deepgram, choose **few more additional ASR systems** to benchmark. Open-source, API-based, or a mix - your call. Justify your picks.

What We're Looking For

We're not looking for a specific answer. We're looking at how you think. Here are some dimensions we care about - you don't need to cover all of them, but the best submissions will touch on several:

Model selection - Which ASR systems did you choose to benchmark against Deepgram and why? Did you consider open-source vs API-based? Multilingual vs language-specific? Different model sizes? What tradeoffs drove your picks?

Metrics - What did you measure? Why those metrics and not others? Are there things that standard metrics miss - especially for entity-heavy utterances like locality names?

Evaluation methodology - How did you slice the data? Did you consider different audio conditions, speaker profiles, utterance lengths, or language mixing patterns? How did your self-recorded samples compare to open-source dataset results (if you used both)?

Beyond transcription accuracy - Things like latency (first-byte, total), throughput, streaming vs batch behavior, memory footprint, ease of deployment, cost at scale. You won't be able to measure all of these rigorously - but acknowledging what matters in production and attempting to measure even a few is valuable.

Failure analysis - Where do models break? Is it noise? Accents? Code-switching? Specific locality names? Short utterances? Show us the failure modes, not just aggregate scores.

Interpretation - Raw numbers are useless without a story. What would you actually recommend and under what constraints?

Deliverables

1. **Audio recordings** - All 20 locality audio files, clearly named.
2. **Code** - A reproducible pipeline (Python preferred) that loads your recordings, runs inference across all models, and computes your chosen metrics. Clean, documented, runnable.
3. **Report** - A written document (max 3 pages, can be markdown) covering: your approach, key findings, failure analysis, and recommendation. All models compared against Deepgram as baseline. Tables and charts welcome. Filler and fluff not welcome.
4. **A 10-minute walkthrough** - Be ready to present your findings and defend your choices. We will ask follow-up questions.

Logistics

- **Time limit:** 5 days from receiving this assignment.
- **Compute:** Use whatever you have access to - Colab free tier is fine. If a model doesn't fit on your hardware, that's a valid data point, not a blocker.
- **Questions:** You can email us clarifying questions, but part of the assessment is figuring out what to do with ambiguity. Don't wait for permission to make reasonable decisions.

What a Strong Submission Looks Like

- Recordings feel real - varied conditions, natural speech, not read off a script
- Benchmarks Deepgram + 2 or more systems with a clear rationale for selection
- Goes beyond WER as the only metric
- Has at least one insight that surprised you
- Acknowledges limitations honestly
- Code runs without hand-holding
- Report is concise and opinionated - tells us what to do, not just what you found

What a Weak Submission Looks Like

- All 20 recordings sound identical - same room, same tone, same sentence structure
 - Only runs Whisper against Deepgram, reports WER, calls it a day
 - Copy-pastes a HuggingFace tutorial without understanding it
 - No failure analysis or qualitative examples
 - Report reads like a college lab manual
 - "I couldn't run X because of compute" with no attempt to work around it
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Good luck. We'd rather see a thoughtful partial analysis than a sloppy complete one.